TIM JOHNSTON

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CURRENT POSITION

CEREMADE, Université Paris-Dauphine — September 2024- Present
Postdoctoral Researcher — *Paris, France*

- Postdoc research under the supervision of Christian Robert
- Developing the mathematics and statistics of privacy
- Numerical analysis of singular stochastic differential equations
- Funded by ERC grant OCEAN https://oceanerc.com/

EDUCATION

The University of Edinburg — September 2020- August 2024
PhD Student in Mathematics — *Edinburgh, Scotland*

- PhD under supervision of Sotirios Sabanis and István Gyöngy
- Viva jury of internal examiner Jiawei Li and external examiner Eric Moulines
- Research consisted of stochastic analysis applied to theoretical and practical problems
- Submitted thesis 'The Euler Scheme for SDEs with Non-Standard Coefficients'

University College London — September 2016- August 2020
Bachelor and Masters Student in Mathematics — *London, England*

- MSci Mathematics
- First Class Honours
- EPSRC funded summer research project in geometry June-August 2019, under the supervision of Costante Bellettini
- Submitted masters thesis 'Curvature Estimates for Weakly Stable Prescribed-Mean-Curvature Hypersurfaces Of Dimension at Most 6'

PAPERS

- **A tamed Euler scheme for SDEs with non-locally integrable drift coefficient**
  Tim Johnston, Sotirios Sabanis
  Stochastic Processes and their Applications. 191 (2026)
- **Differential privacy guarantees of Markov chain Monte Carlo algorithms**
  Andrea Bertazzi, Tim Johnston, Gareth Roberts, Alain Durmus
  ICML 2025
- **The Performance Of The Unadjusted Langevin Algorithm Without Smoothness Assumptions**
  Tim Johnston, Iosif Lytras, Nikos Makras, Sotirios Sabanis
  Transactions on Machine Learning Research (2025)
- **Interacting Particle Langevin Algorithm for Maximum Marginal Likelihood Estimation**
  Ömer Deniz Akyıldız, Francesca Romana Crucinio, Mark Girolami, Tim Johnston, Sotirios Sabanis
  ESAIM: Probability & Statistics 29 (2025)
- **Taming the Interactive Particle Langevin Algorithm - the superlinear case**
  Tim Johnston, Nikos Makras, Sotirios Sabanis
  Applied Mathematics & Optimization. 91 (2025)

- **A Strongly Monotonic Polygonal Euler Scheme**
  Tim Johnston, Sotirios Sabanis
  Journal of Complexity 80 (2024)
- **Kinetic Langevin MCMC Sampling Without Gradient Lipschitz Continuity**
  Tim Johnston, Iosif Lytras, Sotirios Sabanis
  Journal of Complexity 85 (2024)

Pre-Prints

- **Bayesian Adversarial Privacy**
  Cameron Bell, Tim Johnston, Antoine Luciano, Christian Robert
  submitted to *Bayesian Analysis*
- **A Lyapunov-Tamed Euler Method for Singular SDEs**
  Tim Johnston, Pierre Monmarché
  submitted to *SMAI Journal of Computational Mathematics*

TEACHING

Markov Chain Monte Carlo Methods — Semester 1, 2025-2026
Lecturer 3 hours — *Paris Dauphine*

· Delivered lectures on the mixing of Markov chains

Introduction aux probabilités — Semester 2, 2024-2025
Teaching assistant 20 hours, marking 10 hours — *Paris Dauphine*

· Delivered weekly problem classes and marked midterm exam

Applied Stochastic Analysis — Semester 1, 2023-2024
Tutor 30 hours — *The University of Edinburgh*

· Gave weekly tutorials

Stochastic Modelling — Semester 1, 2022-2023
Tutor 30 hours, marking 30 hours — *The University of Edinburgh*

· Gave weekly tutorials and marked weekly problem sheets

Stochastic Analysis in Finance — Semester 1, 2022-2023
Tutor 30 hours, marking 30 hours — *The University of Edinburgh*

· Gave weekly tutorials and marked weekly problem sheets

Honours Complex Variables — Semester 2, 2021-2022
Tutor 30 hours, marking 30 hours — *The University of Edinburgh*

· Gave weekly tutorials and marked weekly problem sheets

SUPERVISION

Summer Project — Summer 2025
Project co-supervisor — *Paris Dauphine*

· Co-supervised visiting student Shlok Mishra on a project relating to the privacy of MCMC releases

TALKS

- **Bayesian Adversarial Privacy**
  SIMPAS Seminar
  Ecole Polytechnique, 25 March 2026
- **Privacy of MCMC**
  Mostly Monte Carlo Seminar
  PariSanté Campus, 17 October 2025
- **Privacy Guarantees for Stochastic Processes**
  OCEAN Retreat
  PariSanté Campus, 3 February 2025
- **Privacy Guarantees for Stochastic Processes**
  North British Probability Seminar
  The University of Edinburgh, 4 March 2025
- **A Pertubation Approach to Differential Privacy**
  Statistical Privacy Research Workshop
  Rocky Pop Hotel Chamonix, 21 June 2025
- **Differential Privacy of Markov Chains**
  GT Markov Working Group
  Sorbonne Université, 13 February 2025
- **An Euler Scheme for SDEs with non-locally integrable coefficients**
  Numerical analysis and applications of SDEs (NAASDE)
  BCIM Bedlewo, 24 September 2024
- **Numerical Methods for SDEs with Irregular Coefficients**
  Mostly Monte Carlo Seminar
  PariSanté Campus, 13 September 2024
- **The Performance of the Euler Scheme for SDEs with Discontinuous Drift**
  International workshop on diffusions in machine learning
  Alan Turing Institute, 15 July 2024
- **The Performance of the Euler Scheme for SDEs with Discontinuous Drift**
  École Polytechnique, 19 February 2024
- **Stochastic Differential Equations with Non-Standard Coefficients and their Approximations**
  Monte Carlo Methods and Applications (MCM)
  Sorbonne Université, 26-30 June 2023
- **Interacting Particle Systems for EM**
  10th International Workshop on Applied Probability
  Thessaloniki, Greece, 7 June 2023
- **Interacting Particle Systems for EM**
  Accelerating generative models and nonconvex optimisation
  The Alan Turing Institute, 24 March 2023

POSTERS

- **Differential Privacy of Markov Chains**
  ICML 2025
  Vancouver, 13-19 July 2025

- **Lyapunov-Tamed Euler Schemes**
  EURANDOM Advances in MCMC methods workshop
  Eindhoven, 10-12 December 2025

ORGANISATION

Mostly Monte Carlo Seminar September 2025 - Present
https://antoineluciano.github.io/mostlymontecarlo/ *PariSanté Campus*

· Booked rooms, invited and coordinated speakers

REFEREEING

- **Journal of Complexity**
  https://www.sciencedirect.com/journal/journal-of-complexity
  Reviewed two papers 2024-2025
- **Stochastic Processes and their Applications**
  https://www.sciencedirect.com/journal/stochastic-processes-and-their-applications
  Reviewed two papers 2024-2026

LANGUAGES

- **English**
  Fluent, maternal
- **French**
  Learning, B1 level

WORK EXPERIENCE

Harrison Allen (Personal Tuition Company) 2018 - 2020
https://www.harrisonallen.co.uk *Personal Tutor in Mathematics*

· Individually planned and executed multiple one-on-one lessons a week

The Triangular Learning Centre (Tuition Centre) 2017 - 2018
https://triangularlearningcentre.co.uk/ *Teaching Assistant in Mathematics*

· Executed sessions for small groups of primary school pupils

The Skill Academy (Tuition Centre) 2017 - 2018
http://www.theskillacademy.org/ *Teaching Assistant in Mathematics*

· Planned and executed lessons to small groups of pupils